

Vaibhav Gollapalli

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

College Station, TX

Expected Graduation: May 2028

- **Relevant Coursework:** Foundations of Computing, Data Structures & Algorithms, Multivariable Calculus, Linear Algebra & Vector Geometry, Discrete Mathematics, Differential Equations, Statistics

EXPERIENCE

Software Engineering Intern

Soularis

May 2026 – Present

Austin, TX

- Built **API** abstraction layers in **TypeScript**, consolidating **10+** backend endpoints into a unified interface.
- Implemented vector search pipeline using **Supabase pgvector**, powering recommendations across **5** categories.
- Automated **CI/CD** via **GitHub Actions** and **Playwright**, orchestrating **670+** runs across production, staging.

Research Fellow

UNT Mixed Realities Lab

Nov 2024 – Present

Denton, TX

- Engineered a **WGAN-GP** model to synthesize **EEG**, matching real transport energy within **1 std. dev.**
- Architected a real-time **EEG** streaming pipeline via **Emotiv EPOC X + MQTT**, achieving **512ms** latency.
- Developed a label-free cognitive load index using **score-based diffusion**, outperforming baselines at **0.613 AUC**.

AI Systems Engineering Intern

CORE (YC S23)

May 2025 – August 2025

Remote

- Designed a distributed **MCP** for AI-driven automation, enabling **multi-agent** architectures with context.
- Optimized backend **APIs** and **system prompts** for AI automation, improving **task success rate** by **30%**.
- Built internal tools for debugging and documentation automation, reducing repetitive engineering tasks by **30%**.

PROJECTS

Mean Eats | Swift, SwiftUI, Supabase, AWS Lambda, BeautifulSoup4

Nov. 2025 – Present

- Shipped App Store dining app to **200+** UNT users across **5** halls with live menus and meal-plan tracking.
- Automated **5** dining scrapers via **AWS Lambda** + EventBridge, syncing daily menus with **180+** days uptime.
- Built offline cache and on-device recommendations, cutting redundant API calls by **60%+** on every launch.

Glyph | Swift, SwiftUI, Core ML, Fireworks AI, PyTorch

Jan. 2026 – Present

- Built iPad handwriting app with **Apple Pencil** and AI-powered circle-to-answer using **Apple Vision OCR**.
- Trained **6.5M**-parameter LSTM+MDN model on **11,739** samples across **1,730** writers from IAM-OnDB.
- Integrated **26-letter** handwriting synthesis engine with **Fireworks AI** LLMs for automated answer generation.

OpenGreyWolfOptimizer | Python, NumPy, mpi4py, pytest

Jul. 2025 – Present

- Shipped open-source GWO package with sklearn-style **.fit()** API, full type hints, and a JOSS paper draft.
- Benchmarked optimizer on **4** functions, reaching **4.4e-16** Sphere and **0.0** Rastrigin fitness in under **0.03s**.
- Added optional **MPI** parallel fitness evaluation plus **13** pytest cases and CI across **3** Python versions.

TECHNICAL SKILLS

Programming: Python, Swift, JavaScript, TypeScript, C++, Java, C, C#, PL/SQL, PostgreSQL, HTML, CSS

Frameworks & Libraries: SwiftUI, React, React Native, Tailwind, FastAPI, Node.js, Express, PyTorch, TensorFlow, NumPy, Pandas

Tools & Platforms: Xcode, Git, Docker, AWS Lambda, Supabase, Firebase, Postman, Figma, Vite, GitHub Actions

AI & ML: Core ML, LSTM, MDN, Fireworks AI, WGAN-GP, scikit-learn, Federated Learning, MQTT, RAG, LangChain

Testing & DevOps: Pytest, Playwright, Unit Testing, CI/CD, Docker, Kubernetes, Redis

Publications:

- “Federated TinyML for Lightweight Anomaly Detection in IoMT Using MQTT-Enabled Edge Deployment”
- “StrokeSentry: A Novel Edge AI-Mobile Application for Rapid Stroke Detection”
- “A Hybrid Windkessel-Neural Approach for Improved Noninvasive Blood Pressure Monitoring”
- “Cognitive Energy Modeling for Neuroadaptive Human-Machine Systems using EEG and WGAN-GP”